

K226 Elie Toy 3862 Substrate Hubble Tension Ratio 12/13 Audit

Keeper (Claude Opus 4.7)

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0. Purpose

K-audit on Elie Toy 3862 substrate prediction $H_0^{\text{Planck}}/H_0^{\text{SHOES}} = (C_2 + g - 1)/(C_2 + g) = 12/13 \approx 0.9231$ for the Hubble tension ratio. Per CI_BOARD Friday K226 priority + Cal #242 STRUCTURAL TENSION reframe explicit.

This K-audit is structurally distinct from K221-K225 because the observational referent itself is contested. K226 identifies and operationalizes a NEW fourth audit category that emerges naturally from H_0 tension structure: STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION.

1. Substrate prediction in operator-language

Substrate prediction (Toy 3862):

$$\frac{H_0^{\text{Planck}}}{H_0^{\text{SHOES}}} = \frac{C_2 + g - 1}{C_2 + g} = \frac{12}{13} = 1 - \frac{1}{13}$$

with $13 = C_2 + g = 6 + 7$ as 2-anchor BST-primary sum, and $\$1 = \$$ substrate identity-element per Casey #5 Integer Web. The $1 - 1/13$ form parallels K225 $n_s = 1 - 1/28$ structure: both are substrate-natural deviations from unity by $1/(\text{BST-primary-composite})$.

Numerical value: $12/13 = 0.923077 \dots$

In operator-language: substrate predicts that CMB-epoch H_0 measurement vs late-time H_0 measurement differ by substrate-natural factor $(C_2 + g - 1)/(C_2 + g)$, interpreted as substrate identity-element subtraction ($1/13$ substrate-shift between epochs).

2. Precision per H₀ source pair (Cal #242 source-pinning operational — STRONG concern)

The substrate prediction is a RATIO of two H₀ measurements. Both measurements have observational uncertainties + contested-source landscape. Computing deviation across major H₀ source pairs:

Source pair	H ₀ Planck	H ₀ late-time	Ratio	Deviation from 12/13
Planck 2018 / SH0ES 2022 (Riess)	67.36	73.04	0.9222	0.093%
Planck PR4 / SH0ES 2022	67.4	73.04	0.9228	0.025%
Planck 2018 / SH0ES 2023	67.36	73.04	0.9222	0.093%
Planck 2018 / TRGB CCHP (Freedman 2024)	67.36	69.8	0.9651	4.55%
DES SN / SH0ES	67.8	73.04	0.9282	0.55%
Planck 2018 / JWST recent (mixed)	67.36	~70-73	0.923-0.962	0.06-4.2%

The substrate prediction $12/13 = 0.9231$ sits at ~0.1% precision when using Planck 2018 + SH0ES 2022 Cepheid-SN reference pair, but deviates substantially for TRGB-based late-time H₀ measurements (~4-5%).

The choice of source pair determines whether substrate prediction matches. This is a much stronger Cal #242 source-pinning concern than K224 λ_H or K225 n_s.

Cal #242 source-pinning verdict (REVISED for STRUCTURAL TENSION reframe):

Substrate predicts EXACT ratio $(C_2 + g - 1)/(C_2 + g) = 12/13$ for CMB-epoch / late-time H₀ measurement pair, with substrate-mechanism via identity-element subtraction per Casey #5 Integer Web. Precision claim 0.09% holds ONLY at Planck 2018 / SH0ES 2022 source pair selection; alternative late-time H₀ measurements (TRGB, DES SN, JWST) yield 0.55-4.55% deviation. Honest framing requires explicit (a) source-pair specification, (b) acknowledgment that H₀ tension itself is observationally contested, (c) substrate prediction is CONDITIONAL on the tension being real and resolving to Cepheid-SN late-time H₀ measurement.

3. The STRUCTURAL TENSION reframe (Cal #242 explicit)

K226 substrate prediction differs structurally from K221-K225 substrate predictions because:

K221-K225 substrate predictions: substrate predicts a single observable value (e.g. $\sin^2 \theta_{13}$, θ_{QCD} , r , λ_H , n_s); observed value is a single PDG-precision quantity (with source-uncertainty per Cal #242); audit compares substrate prediction to observed value.

K226 substrate prediction: substrate predicts that TWO DIFFERENT measurement methods will yield TWO DIFFERENT H_0 values in a specific RATIO 12/13. This is a META-prediction about an observational anomaly structure.

If H_0 tension turns out to be observational systematics (e.g., JWST Cepheid calibration corrections eliminate the tension), substrate prediction of 12/13 ratio becomes either (a) IRRELEVANT (no tension to predict) or (b) FALSIFIED (substrate predicted a tension that doesn't exist).

This is a NEW audit category that K226 operationalizes: STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION.

Category	Definition	Example
Tier 1 EXACT MATCH	Substrate matches observed PDG-precision	$\sin^2 \theta_{13}$, m_τ/m_e
EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION	Substrate predicts EXACT value satisfying observational bound at source-pinned precision	$\theta_{\text{QCD}} = 0$ (K222), $\lambda_H = 4/31$ (K224), $n_s = 27/28$ (K225)
FALSIFIER-DRIVEN PREDICTION	Substrate predicts specific value within current bound; future experiment tests	$r \approx \alpha^2$ (K223), n_s secondary, m_{Planck}/m_e (K227 pending)
STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION (NEW per K226)	Substrate predicts the EXISTENCE and STRUCTURE of an observational anomaly (not just a single value); conditional on the anomaly being real	H_0 ratio 12/13 (K226)
Casey #5 Integer Web multi-source	Same observable value via multiple substrate-mechanism paths	207 (Lyra Composite + Elie $N_{\text{max}}+70$), 8/7 (2-cluster)

4. Per-gate analysis

Gate G1: Hubble tension observational — PASS with STRUCTURAL caveat

Toy correctly states Planck 2018 67.36 ± 0.54 and SH0ES 2022 73.04 ± 1.04 with “ 5σ tension.” But omits the contested-source landscape: TRGB CCHP gives 69.8; JWST recent results mixed; H_0 tension may be smaller or non-existent depending on late-time methodology choice.

PASS with caveat: H_0 tension is OBSERVATIONALLY CONTESTED, not settled. Substrate prediction of 12/13 ratio is CONDITIONAL on tension being real with Cepheid-SN late-time methodology being correct.

Gate G2: Substrate ratio = 12/13 — PASS at substrate-natural-form IDENTIFICATION

Substrate form 12/13 is substrate-natural at the BST-integer level: $-13 = C_2 + g$ is a 2-anchor BST-primary sum $-12 = 13 - 1 = C_2 + g - 1$ via substrate identity-element subtraction (Casey #5 Integer Web) - Alternative readings: $12 = \text{rank} \cdot C_2 = (N_c + 1) \cdot N_c = 2g - 2$ - Structural parallel to K225 $n_s = 1 - 1/28$: both are substrate-natural $1 - 1/(\text{BST-primary-composite})$ forms

PASS at Cal #34 substrate-natural-form IDENTIFICATION. Substrate value 12/13 is BST-integer-derived.

Gate G3: Substrate-mechanism via substrate Integer Web identity-element — CONDITIONAL with REVISION

Toy claims substrate-mechanism via “substrate identity + $(C_2 + g)$ ” interpretation: “Local H_0 (SH0ES) measures dimensionless $1 = \text{substrate identity}$; CMB H_0 (Planck) measures $(C_2 + g - 1)/(C_2 + g) = 12/13$ substrate-shifted.”

This claim REQUIRES substantive revision:

1. The interpretation that SH0ES measures “ $1 = \text{substrate identity}$ ” and Planck measures “12/13” assumes a specific assignment of which H_0 corresponds to which substrate value. The substrate framework does NOT a priori distinguish CMB-epoch from late-time epoch as “shifted vs unshifted” — both are physical measurements of expansion rate.
2. Per Cal #189 substrate-mechanism FORWARD vs substrate-natural-form IDENTIFICATION distinction: this is substrate-natural-form IDENTIFICATION of the RATIO, NOT substrate-mechanism FORWARD derivation of WHY one measurement should equal substrate value v and another $v \cdot 12/13$.
3. The framing “This SOLVES the Hubble tension as substrate-natural feature” is OVERCLAIM. Substrate identifies a substrate-natural ratio FORM 12/13 that matches the observed tension at Planck 2018 + SH0ES 2022 source-pair selection. Substrate does NOT derive the substrate-mechanism for WHY the

CMB-epoch measurement should differ from the late-time measurement by exactly 1/13.

CONDITIONAL PASS at substrate-mechanism FRAMEWORK level, with substantive revision needed in claim language. Multi-week substrate-mechanism FORWARD derivation required: substrate τ -evolution between CMB-epoch and late-time-epoch must be shown to FORCE the 12/13 ratio via substrate-Bergman heat-flow at operator level.

Gate G4: Cross-link to substrate-cosmology primitive — PASS with cluster identification

Toy catalogs substrate-cosmology primitive 7 readings (Λ , Σm_ν , n_s , H_0 ratio, cosmogony, Interstasis cyclic). Per Cal #36 STANDING positive-search: substrate-cosmology primitive cluster is a substrate-Schur-generator candidate at COSMOLOGY sector.

Substrate-cosmology Cat A primitive cluster identification (cross-link K223 + K225 + K226): - K223 $r \approx \alpha^2$ (FALSIFIER-DRIVEN, inflation) - K225 $n_s = 27/28$ (EXACT-BOUND-SATISFYING + FALSIFIER-DRIVEN, inflation) - K226 H_0 ratio 12/13 (STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION, late-universe)

Per Grace G14 R-4/R-5 K-type-AGGREGATE methodology + Cal #35 STANDING brake on multi-source convergence: these form a substrate-cosmology Cat A primitive cluster with 3 readings. Per Cal #36 STANDING positive-search: 1 effective independent cluster-level contribution (NOT 3 independent contributions).

Gate G5: Honest tier verdict — REVISED per K226 STRUCTURAL TENSION re-frame + Cal #242 + Cal #189 + Grace G14

Toy claims “TIER 1 EXACT CANDIDATE at 0.09%” and “HUBBLE TENSION substrate-mechanism candidate” and “This SOLVES the Hubble tension as substrate-natural feature.”

Revised per K226 audit:

- Substrate prediction 12/13 = 0.923077 EXACT at substrate-natural-form IDENTIFICATION (Cal #34 PASS)
- Observed ratio depends on H_0 source-pair selection: Planck 2018 / SH0ES 2022 gives 0.09% deviation; TRGB-based or DES SN-based alternatives give 0.55-4.55% deviation
- Per Cal #242 source-pinning: precision claim must pin BOTH H_0 sources explicitly; honest framing “0.09% deviation at Planck 2018 / SH0ES 2022 selection; 0.55-4.55% at alternative late-time source pairs”
- Per Cal #189 substrate-mechanism FORWARD vs IDENTIFICATION: this is RATIO-FORM IDENTIFICATION, NOT mechanism-FORWARD derivation

- Per Grace G14/G15: $13 = C_2 + g$ 2-anchor BST-primary sum + identity-element subtraction; 2-cluster CROSS-CLUSTER Casey #5 Integer Web instance, NOT single-cluster substrate-Schur-generator
- NEW STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION audit category: substrate predicts EXISTENCE and STRUCTURE of an observational anomaly; CONDITIONAL on anomaly being real
- “SOLVES the Hubble tension” claim retracted: substrate IDENTIFIES a substrate-natural ratio matching observed tension at specific source-pair; substrate does NOT close substrate-mechanism FORWARD derivation closing why the tension should exist

5. K226 verdict

PASS at FRAMEWORK tier per Cal #34 substrate-natural-form IDENTIFICATION + Cal #36 STANDING positive-search cascade (substrate-cosmology primitive 7 readings + substrate-cosmology Cat A primitive cluster identification with K223 + K225).

Audit category: STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION (NEW, K226 operationalized) — substrate predicts EXISTENCE and STRUCTURE of an observational anomaly (12/13 ratio between CMB-epoch and late-time epoch H_0 measurements); CONDITIONAL on H_0 tension being real AND on Cepheid-SN late-time H_0 methodology being correct.

CONDITIONAL on: - Observational settlement of H_0 tension: if JWST + TRGB systematic improvements eliminate the tension, K226 substrate prediction is FALSIFIED (substrate predicted a tension that doesn't exist). Current contested-source landscape gates the audit. - Cal #189 multi-week substrate-mechanism FORWARD derivation: substrate τ -evolution between CMB-epoch and late-time-epoch must derive 12/13 ratio via substrate-Bergman heat-flow at operator level; without this, claim is substrate-natural-form IDENTIFICATION not substrate-mechanism FORCING - Cal #242 source-pinning: claims pin Planck data release + SH0ES 2022 explicitly; honest framing 0.09% AT Planck 2018 / SH0ES 2022 source pair, 0.55-4.55% at alternative source pairs - Cal #35 STANDING brake on “SOLVES the Hubble tension” framing: revise to “substrate IDENTIFIES substrate-natural 12/13 form matching observed ratio at specific source-pair selection”

Cumulative effect on “11 Tier 1 EXACT” → effective independent count:

Per K221-K226 cascade + Lyra F22 v1.7 + Grace G14/G15: - Substrate H_0 ratio 12/13 enters at STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION category (NEW; structurally distinct from K221-K225 categories) - 2-cluster CROSS-CLUSTER Casey #5 Integer Web instance ($C_2 + g$ sum + identity-element subtraction) - Substrate-mechanism FORWARD CONDITIONAL per Cal #189 multi-week - CONDITIONAL on tension being real: STRUCTURAL TENSION reframe per Cal #242 — observational referent itself contested

Effective contribution to cumulative independent count: K223 + K225 + K226

jointly form substrate-cosmology Cat A primitive cluster (1 effective combined contribution), NOT 3 separate Tier 1 candidates. K226 does not increment count separately from K223+K225 substrate-cosmology cluster.

Running cumulative honest framing post-K226: - K222 Strong CP $\theta_{\text{QCD}} = 0$: 1.0 effective (substrate-CP preservation) - K223+K225+K226 substrate-cosmology Cat A cluster: 1.0 effective combined - K224 $\lambda_{\text{H}} = 4/31$: 0.5 effective contribution - Total Friday K-audit cascade contribution: ~2.5 effective independent substrate-mechanism FORCING candidates added to cumulative count

Running total cumulative ~6-7 effective independent substrate-mechanism FORCING candidates post-K226 (consistent with K224 ~6.5 + K225/K226 cluster identification keeping count flat).

6. NEW four-category audit framework operational

K226 operationalizes the NEW fourth audit category:

STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION: substrate predicts the EXISTENCE and STRUCTURE of an observational anomaly (not just a single observable value); CONDITIONAL on the anomaly being real (not observational systematics). Falsifier: if the observational anomaly is resolved by improved measurement methodology (e.g., JWST Cepheid recalibration eliminates H_0 tension), substrate prediction is FALSIFIED.

This category is structurally distinct from: - Tier 1 EXACT MATCH (substrate matches single observed value at PDG precision) - EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION (substrate predicts EXACT value within source-uncertainty envelope) - FALSIFIER-DRIVEN PREDICTION (substrate predicts value within current bound; future experiment tests at higher precision)

Substrate predictions in STRUCTURAL-TENSION category from BST corpus: - H_0 ratio 12/13 (K226) — primary instance - Potentially: ε'/ε direct vs indirect CP-violation tension (Five-Absence v0.4 territory) - Potentially: muon g-2 anomaly direction - Future identifications welcome

Operational discipline for this category: distinguish (a) substrate IDENTIFIES a tension-natural ratio FORM from (b) substrate DERIVES the tension structure from mechanism. (a) is Cal #34 substrate-natural-form IDENTIFICATION; (b) requires Cal #189 substrate-mechanism FORWARD derivation.

7. Cross-CI three-granularity integration

Granularity	Disposition for H_0 ratio = 12/13
Keeper 3+1-category audit (observable-prediction)	STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION primary (NEW); CONDITIONAL on H_0 tension being real + Cal #189 substrate-mechanism FORWARD
Grace 11-cluster taxonomy (substrate-anchor)	2-cluster CROSS-CLUSTER Casey #5 Integer Web instance: Cluster A Casey #5 sum ($C_2 + g$) + Cluster A Casey #5 identity-element subtraction (1)
Lyra substrate-K-type $\times$ $SU(N_c)$ tensor product	substrate τ -evolution between CMB-epoch (Bergman heat-flow at large τ) and late-time epoch (boundary substrate K-type ground state); $SU(N_c)$ singlet; substrate-cosmology scalar sector

Three-granularity convergence: H_0 ratio 12/13 is substrate-natural at multiple resolutions; substrate-mechanism FORWARD CONDITIONAL at all three resolutions; STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION primary category operational at observable-prediction granularity.

8. Recommendations

1. Toy 3862 v0.2 absorption: pin Planck data release + SH0ES year explicitly (“substrate 12/13 EXACT at Planck 2018 / SH0ES 2022 source-pair; 0.09% at this pair; 0.55-4.55% at alternative late-time H_0 source pairs including TRGB, DES SN, recent JWST”); state CONDITIONAL on H_0 tension being real (current contested-source landscape); state CONDITIONAL on Cal #189 multi-week substrate-mechanism FORWARD derivation; retract “SOLVES the Hubble tension” claim — replace with “substrate IDENTIFIES substrate-natural ratio FORM matching observed tension at specific source-pair selection.”
2. Multi-week substrate-mechanism FORWARD derivation prerequisite: substrate τ -evolution between CMB-epoch and late-time-epoch must derive 12/13 ratio at operator level. Cross-link to Lyra F2 substrate-cognition Phase 1 (substrate τ -evolution structure) + Toy 3787 substrate cosmogony origin framework + Casey #14 STANDING 3+1 Minkowski.
3. K223 + K225 + K226 substrate-cosmology Cat A cluster identification: file as cluster-level entry (1 effective combined contribution), NOT 3 separate Tier 1 candidates. Aligns with Grace G14 R-4/R-5 K-type-AGGREGATE methodology + Cal #35 STANDING brake.

4. NEW four-category audit framework: extend three-category framework (K222 + K223 + K224 + K225) to four categories adding STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION. Document in Methodology Index v0.15+ via Cal cold-read.
5. External-facing claims: Cal #50 STANDING DOUBLE-LOCKED EXTERNAL applies — “substrate SOLVES Hubble tension” claim NOT externally usable; “substrate IDENTIFIES substrate-natural ratio FORM matching observed tension structure” honest framing for internal use.
6. K-audit cascade continuation: K227 (m_{Planck}/m_e Tier 1 cross-anchor) + α^{-1} BORDERLINE next at proper depth.

9. Closure

K226 Elie Toy 3862 substrate $H_0^{\text{Planck}}/H_0^{\text{SH0ES}} = (C_2 + g - 1)/(C_2 + g) = 12/13$:

PASS at FRAMEWORK tier + STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION audit category (NEW, K226 operationalized) + CONDITIONAL on (a) H_0 tension being real with Cepheid-SN late-time methodology + (b) Cal #189 multi-week substrate-mechanism FORWARD derivation + (c) Cal #242 source-pair pinning + (d) Cal #35 STANDING “SOLVES” overclaim retraction.

Honest precision framing: 0.09% deviation at Planck 2018 / SH0ES 2022 source pair; 0.55-4.55% at alternative late-time H_0 source pairs. Substrate IDENTIFIES substrate-natural 12/13 form matching observed tension; substrate does NOT derive substrate-mechanism FORWARD for tension existence.

NEW four-category audit framework operational: Tier 1 EXACT MATCH + EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION + FALSIFIER-DRIVEN PREDICTION + STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION.

Cumulative inventory: K223 + K225 + K226 jointly form substrate-cosmology Cat A primitive cluster (1 effective combined contribution per Grace G14 R-4/R-5 K-type-AGGREGATE methodology + Cal #35 STANDING brake). K226 does not increment count separately from cluster.

— Keeper, Friday 2026-06-05 ~15:00 EDT. K226 Elie Toy 3862 substrate H_0 ratio 12/13 PASS at FRAMEWORK + NEW STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION audit category (operationalized this audit) + CONDITIONAL on H_0 tension being observationally real + Cal #189 substrate-mechanism FORWARD derivation + Cal #242 source-pair pinning + Cal #35 “SOLVES” overclaim retraction. Honest precision 0.09% at Planck 2018 / SH0ES 2022 source pair; 0.55-4.55% at alternative late-time H_0 source pairs. Substrate-cosmology Cat A primitive cluster identification (K223+K225+K226) = 1 effective combined contribution NOT 3 separate. Four-category audit framework operational + extended to Methodology Index v0.15 via Cal cold-read.